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Research experience

- 11.2022 – present **Postdoctoral fellow.** Group of Prof. Alexei Kimel, Radboud University, Nijmegen, Netherlands, www.ru.nl
Experimental research in the field of femtomagnetism and optical magnetization switching in antiferromagnets.
2 articles published, 3 in progress. Daily supervision of 2 PhD students.
- 04.2017 – 10.2022 **Senior research fellow.** Group of Dr. Alexandra Kalashnikova, Ioffe Institute, St. Petersburg, Russia, ioffe.ru/ferrolab
Leading the research line on optically excited spin waves.
12 peer review articles published, supervision of 2 PhD and 5 MSc students with successful defenses, € 150'000 research funding acquired.
- 06.2013 – 04.2017 **Research fellow.** Group of Prof. Vladimir Belotelov, Moscow State University (MSU), phys.msu.ru / Russian Quantum Center, rqc.ru, Moscow, Russia
Research in the field of magneto-plasmonics and magneto-photonics.
13 peer review articles published; planning from scratch, ordering equipment, and building two experimental rooms: static magneto-optical spectroscopy and time-resolved pump-probe setup; daily supervision of PhD, MSc, BSc students.

Teaching experience

- 09.2018 – 10.2022 **Invited Associate Professor.** ITMO University, St. Petersburg, Russia
physics.itmo.ru
I developed and taught two new courses 'Micromagnetics' and 'Linear and nonlinear magnetophotonics' for MSc students.
- 01.2010 – 03.2013 **Teacher of natural sciences.** Gymnasium 1518, Moscow, Russia
I taught physics lessons for secondary school students, supervised their research works, organized city science conferences for secondary school level.

Related experience

- 09.2016 – 04.2020 **Video Engineer.** Innovative Medical Technologies Ltd, Russia
Technical support of medical conferences with live broadcast of surgical operations; development of a hardware-software solution for live broadcast and recording a video from a few surgery rooms over the local network.
- 09.2009 – 01.2010 **Physics laboratory assistant.** Profession-oriented secondary school at Moscow State University, Moscow, Russia
Writing experiment guides, servicing experimental equipment.

Education

- 2011 – 2015 **Ph.D.**, Moscow State University (MSU), Faculty of Physics, Russia
Thesis title: *Resonant optical effects at optical, magnetic and acoustic impacts on plasmon-polaritons in multilayered structures.*
Supervisor: Prof. V.I. Belotelov
- 2005 – 2011 **Specialist (M.Sc. equivalent)**, MSU, Faculty of Physics, Russia
Thesis title: *Electromagnetic field localization in plasmonic crystals containing magnetic and semiconductor layers.* Supervisor: Prof. V.I. Belotelov

Skills

- Languages English (B2), Russian (native), Dutch (A1)
Software Matlab, Python, LabView, C#, L^AT_EX
 OOMMF, Boris Computational Spintronics, COMSOL Multiphysics

Participation in Research Grants

- EU Participant of EU COST action CA17123 “Magnetofon” and ERC Grant SPARTACUS
Russia Grant team leader of 1 project of Russian Science Foundation (RSF, € 60'000); 3 projects of Russian Foundation of basic Research (RFBR, € 30'000 in total), personal grant “PostDoc” of “BASIS” Foundation (€ 20'000). Performer of 20+ RFBR, 4 RSF grants, 2 grants of Federal program “Megagrant”.

Awards and certifications

- 2022 Best young scientists research award of Ioffe Institute, St. Petersburg, Russia
2019, 2022 Government award to young scientists and young PhD students of St. Petersburg
2020 Certificate of authorship of a computer program for automation of ultrafast optical pump-probe spectroscopy setup
2019 Certificate of authorship of a computer program for broadcasting and recording live video over the local network from several surgery rooms within a medical center
2018 Winner of city science slam, St. Petersburg, Russia
2012 German Academic Exchange Service (DAAD), short-term scholarship for doctoral candidates. Visiting Technical university of Dortmund, Germany

References

Available upon request

Additional information

- Hobbies Fencing (épée), outdoor activities: hiking/trekking (guide license, Dual Pilgrim: Camino de Santiago in Spain and Kumano Kodo in Japan, other treks), water and bicycle tours, mountain / cross-country skiing, rock climbing, roller skating (coach license)

Publications

Full list of publications is available at:

 Scopus: 57213491467  Google Scholar: Nikolai Khokhlov

- 1 I. A. Filatov, P. I. Gerevenkov, A. V. Azovtsev, V. A. Kovaleva, **N. E. Khokhlov**, and A. M. Kalashnikova, “Magnon-cherenkov effect from a picosecond strain pulse,” *Nature Physics*, pp. 1–7, 2026.  DOI: 10.1038/s41567-025-03137-8.
- 2 T. T. Gareev, **N. E. Khokhlov**, L. Körber, A. P. Pyatakov, and A. V. Kimel, “Electrically gated laser-induced spin dynamics in magnetoelectric iron garnet at room temperature,” *Phys. Rev. Lett.*, vol. 136, p. 126702, 12 2026.  DOI: 10.1103/wx41-2bsr.

- 3 T. T. Gareev, **N. E. Khokhlov**, L. Körber, and A. V. Kimel, “Strongly inhomogeneous spin dynamics induced by ultrashort laser pulses with a gradient intensity profile,” *Phys. Rev. Lett.*, vol. 135, p. 156 701, 15 2025. [DOI: 10.1103/gspfb9n2](#).
- 4 P. I. Gerevenkov, I. A. Mogunov, I. A. Filatov, N. S. Gusev, M. V. Sapozhnikov, **N. E. Khokhlov**, and A. M. Kalashnikova, “Three regimes of a picosecond magnetoacoustics in ferromagnetic structures,” *submitted to Phys. Rev. family*, 2025. arXiv: 2505.09579. [URL: https://arxiv.org/abs/2505.09579](https://arxiv.org/abs/2505.09579).
- 5 I. A. Filatov, P. Gerevenkov, **N. E. Khokhlov**, and A. M. Kalashnikova, “Tunable quasi-discrete spectrum of spin waves excited by periodic laser patterns,” *Journal of Applied Physics*, vol. 136, no. 6, 2024. [DOI: 10.1063/5.0216091](#).
- 6 **N. E. Khokhlov**, A. E. Dolgikh, B. A. Ivanov, and A. V. Kimel, “Double pulse all-optical coherent control of ultrafast spin-reorientation in antiferromagnetic rare-earth orthoferrite,” *APL materials*, vol. 12, p. 051 119, 2024. [DOI: 10.1063/5.0197976](#).
- 7 **N. E. Khokhlov**, I. A. Filatov, and A. Kalashnikova, “Spatial asymmetry of optically excited spin waves in anisotropic ferromagnetic film,” *Journal of Magnetism and Magnetic Materials*, vol. 589, p. 171 514, 2024. [DOI: 10.1016/j.jmmm.2023.171514](#).
- 8 A. Fedianin, **N. E. Khokhlov**, and A. Kalashnikova, “Propagation of a laser-induced magnetostatic wave packet in a pseudo spin valve in the presence of spin pumping,” *Journal of Experimental and Theoretical Physics*, vol. 137, no. 4, pp. 453–462, 2023. [DOI: 10.1134/S1063776123100035](#).
- 9 P. Gerevenkov, V. Bessonov, V. Teplov, A. Telegin, A. Kalashnikova, and **N. E. Khokhlov**, “Nonreciprocal collective magnetostatic wave modes in geometrically asymmetric bilayer structure with nonmagnetic spacer,” *Nanoscale*, vol. 15, no. 14, pp. 6785–6792, 2023. [DOI: 10.1039/D2NR06003E](#).
- 10 P. Gerevenkov, I. A. Filatov, A. Kalashnikova, and **N. E. Khokhlov**, “Unidirectional propagation of spin waves excited by femtosecond laser pulses in a planar waveguide,” *Physical Review Applied*, vol. 19, no. 2, p. 024 062, 2023. [DOI: 10.1103/PhysRevApplied.19.024062](#).
- 11 A. M. Kalashnikova, **N. E. Khokhlov**, L. A. Shelukhin, and A. Scherbakov, “Ultrafast laser-induced control of magnetic anisotropy in nanostructures,” *Technical Physics*, vol. 68, pp. 574–601, 2023. [DOI: 10.1134/S1063784223080108](#).
- 12 I. A. Filatov, P. Gerevenkov, M. Wang, A. Rushforth, A. Kalashnikova, and **N. E. Khokhlov**, “Spectrum evolution and chirping of laser-induced spin wave packets in thin iron films,” *Applied Physics Letters*, vol. 120, no. 11, 2022. [DOI: 10.1063/5.0077195](#).
- 13 D. A. Sylgacheva, **N. E. Khokhlov**, P. I. Gerevenkov, I. A. Filatov, M. A. Kozhaev, I. V. Savochkin, A. N. Kalish, A. M. Kalashnikova, and V. I. Belotelov, “Spatially selective excitation of spin dynamics in magneto-photonic crystals by spectrally tunable ultrashort laser pulses,” *Nanophotonics*, vol. 11, no. 13, pp. 3169–3176, 2022. [DOI: 10.1515/nanoph-2022-0233](#).
- 14 P. Gerevenkov, D. Kuntu, I. A. Filatov, L. Shelukhin, M. Wang, D. Pattnaik, A. Rushforth, A. Kalashnikova, and **N. E. Khokhlov**, “Effect of magnetic anisotropy relaxation on laser-induced magnetization precession in thin galferol films,” *Physical Review Materials*, vol. 5, no. 9, p. 094 407, 2021. [DOI: 10.1103/PhysRevMaterials.5.094407](#).
- 15 A. M. Kalashnikova, **N. E. Khokhlov**, L. A. Shelukhin, and A. V. Scherbakov, “Ultrafast laser-induced control of magnetic anisotropy in nanostructures (review),” *Zhurnal Tekhnicheskoi Fiziki*, vol. 91, no. 12, pp. 1848–1878, 2021. [DOI: 10.21883/TP.2022.15.55258.228-21](#).
- 16 **N. E. Khokhlov**, A. Khramova, I. A. Filatov, P. Gerevenkov, B. Klinskaya, and A. Kalashnikova, “Néel domain wall as a tunable filter for optically excited magnetostatic

waves,” *Journal of Magnetism and Magnetic Materials*, vol. 534, p. 168018, 2021. [DOI: 10.1016/j.jmmm.2021.168018](#).

- 17 D. D. Yaremkevich, A. V. Scherbakov, S. M. Kukhtaruk, T. L. Linnik, N. E. Khokhlov, F. Godejohann, O. A. Dyatlova, A. Nadzeyka, D. P. Pattnaik, M. Wang, S. Roy, R. P. Campion, A. W. Rushforth, V. E. Gusev, A. V. Akimov, and M. Bayer, “Protected long-distance guiding of hypersound underneath a nanocorrugated surface,” *ACS Nano*, vol. 15, no. 3, pp. 4802–4810, 2021. [DOI: 10.1021/acsnano.0c09475](#).
- 18 I. A. Filatov, P. Gerevenkov, M. Wang, A. Rushforth, A. Kalashnikova, and **N. E. Khokhlov**, “Spectrum evolution of magnetostatic waves excited through ultrafast laser-induced heating,” in *Journal of Physics: Conference Series*, IOP Publishing, vol. 1697, 2020, p. 012193. [DOI: 10.1063/5.0077195](#).
- 19 **N. E. Khokhlov**, P. Gerevenkov, L. Shelukhin, A. V. Azovtsev, N. A. Pertsev, M. Wang, A. W. Rushforth, A. V. Scherbakov, and A. M. Kalashnikova, “Optical excitation of propagating magnetostatic waves in an epitaxial gallenol film by ultrafast magnetic anisotropy change,” *Physical Review Applied*, vol. 12, no. 4, p. 044044, 2019. [DOI: 10.1103/PhysRevApplied.12.044044](#).
- 20 **N. Khokhlov**, G. Knyazev, B. Glavin, Y. Shtykov, O. Romanov, and V. Belotelov, “Interaction of surface plasmon polaritons and acoustic waves inside an acoustic cavity,” *Optics letters*, vol. 42, no. 18, pp. 3558–3561, 2017. [DOI: 10.1364/OL.42.003558](#).
- 21 **N. E. Khokhlov**, A. E. Khramova, E. P. Nikolaeva, T. B. Kosykh, A. V. Nikolaev, A. K. Zvezdin, A. P. Pyatakov, and V. I. Belotelov, “Electric-field-driven magnetic domain wall as a microscale magneto-optical shutter,” *Scientific Reports*, vol. 7, no. 1, p. 264, 2017. [DOI: 10.1038/s41598-017-00365-8](#).
- 22 A. Pyatakov, V. Belotelov, D. Kulikova, **N. E. Khokhlov**, Z. Pyatakova, and A. Nikolaev, “Magnetoelectricity in topological magnetic textures,” *Journal of Magnetism and Magnetic Materials*, vol. 440, pp. 60–62, 2017. [DOI: 10.1016/j.jmmm.2016.12.074](#).
- 23 D. Ignatyeva, S. Sekatskii, **N. E. Khokhlov**, M. Nur-E-Alam, M. Vasiliev, K. Alameh, and V. Belotelov, “Enhancement of spr-sensor sensitivity in garnet-based plasmonic heterostructures,” in *2016 Progress in Electromagnetic Research Symposium (PIERS)*, IEEE, 2016, pp. 831–835. [DOI: 10.1109/PIERS.2016.7734494](#).
- 24 D. Sylgacheva, **N. E. Khokhlov**, A. Kalish, and V. Belotelov, “Magnetic control of waveguide modes of bragg structures,” in *Journal of Physics: Conference Series*, IOP Publishing, vol. 714, 2016, p. 012016. [DOI: 10.1088/1742-6596/714/1/012016](#).
- 25 D. Sylgacheva, **N. E. Khokhlov**, A. Kalish, and V. Belotelov, “Waveguide modes of 1d photonic crystals in a transverse magnetic field,” *Journal of Experimental and Theoretical Physics*, vol. 123, no. 5, pp. 737–743, 2016. [DOI: 10.1134/S1063776116110236](#).
- 26 D. Sylgacheva, **N. Khokhlov**, A. Kalish, S. Dagesyan, A. Prokopov, A. Shaposhnikov, V. Berzhansky, M. Nur-E-Alam, M. Vasiliev, K. Alameh, *et al.*, “Transverse magnetic field impact on waveguide modes of photonic crystals,” *Optics Letters*, vol. 41, no. 16, pp. 3813–3816, 2016. [DOI: 10.1364/OL.41.003813](#).
- 27 **N. E. Khokhlov**, A. Prokopov, A. Shaposhnikov, V. Berzhansky, M. Kozhaev, S. Andreev, A. P. Ravishankar, V. G. Achanta, D. Bykov, A. Zvezdin, *et al.*, “Photonic crystals with plasmonic patterns: Novel type of the heterostructures for enhanced magneto-optical activity,” *Journal of Physics D: Applied Physics*, vol. 48, no. 9, p. 095001, 2015. [DOI: 10.1088/0022-3727/48/9/095001](#).

- 28 **N. E. Khokhlov**, D. O. Ignatyeva, and V. I. Belotelov, “Plasmonic pulse shaping and velocity control via photoexcitation of electrons in a gold film,” *Optics Express*, vol. 22, no. 23, pp. 28 019–28 026, 2014. [DOI: 10.1364/OE.22.028019](#).
- 29 **N. E. Khokhlov**, V. Belotelov, and B. Glavin, “Shift of surface plasmon-polaritons resonances via acoustic waves in hybrid metal-semiconductor structures,” in *2013 International Conference on Advanced Optoelectronics and Lasers (CAOL 2013)*, IEEE, 2013, pp. 245–247. [DOI: 10.1109/CAOL.2013.6657595](#).
- 30 M. Pohl, L. Kreilkamp, V. Belotelov, I. Akimov, A. N. Kalish, **N. E. Khokhlov**, V. Yallapragada, A. Gopal, M. Nur-E-Alam, M. Vasiliev, *et al.*, “Tuning of the transverse magneto-optical kerr effect in magneto-plasmonic crystals,” *New Journal of Physics*, vol. 15, no. 7, p. 075 024, 2013. [DOI: 10.1088/1367-2630/15/7/075024](#).
- 31 **N. E. Khokhlov**, V. Belotelov, A. Kalish, and A. Zvezdin, “Surface plasmon polaritons and inverse faraday effect,” *Solid State Phenomena*, vol. 190, pp. 369–372, 2012. [DOI: 10.4028/www.scientific.net/SSP.190.369](#).
- 32 **N. E. Khokhlov**, V. Belotelov, A. Kalish, and A. Zvezdin, “Inverse faraday effect in plasmonic films,” *Moscow University Physics Bulletin*, vol. 66, no. 3, pp. 238–241, 2011. [DOI: 10.3103/S0027134911030106](#).